DT081A - Signal and Image Processing

Lab 3 Report — Image Processing

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24 October 2026

1 Introduction

This lab covered pixel transforms, colour reduction, filtering, Fourier analysis, histograms and morphology, comparing image changes, time and memory use.

2 Experimental Setup

The experiments used Python, NumPy, OpenCV and Matplotlib, with the 512×512 `peppers.tiff` test image. BGR was converted to RGB for display, with grayscale and synthetic images used where needed.

3 Results and Analysis

3.1 Gamma Correction

Q1.1. The formula was $255(I/255)^{\gamma}$. At $\gamma = 2.0$, Table 1 shows darker outputs; `uint8` conversion drops fractional parts.

Table 1: Gamma calculation, $\gamma = 2.0$.

Input	Before integer conversion	`uint8` output
64	16.063	16
128	64.251	64
192	144.565	144

Q1.2. With $\gamma = 2.2$, mean intensity fell from 120.22 to 61.00. The image darkened and its histogram moved lower, with gaps from integer conversion (Figures 3, left, and 6, left). This nonlinear change relates to unequal visual sensitivity, but is not an exact logarithmic eye model. The lab's γ was used, not camera pre-correction's $1/\gamma$.

3.2 Colorspaces

Q2.1. OpenCV's order is Y, Cr, Cb, with $Y \approx 0.299R + 0.587G + 0.114B$. All channels started at 0; Y/R/G/B maxima were 228/231/237/229, and standard deviations 53.88/45.17/75.05/44.29. G was widest; Y lay between G and R/B. R had many high values and B mostly low ones. Y simplifies brightness-edge detection to one channel, but can miss equal-brightness colour boundaries.

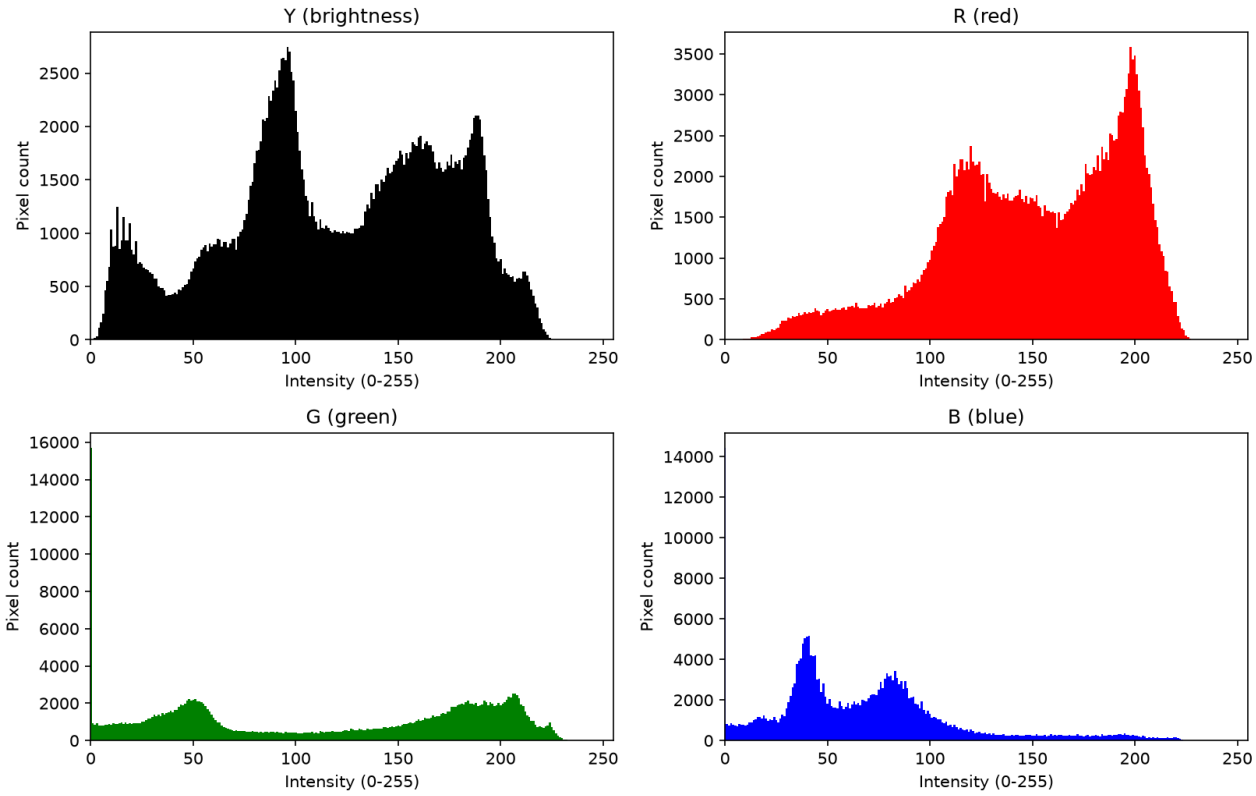


Figure 1: Histograms of the Y (brightness), R, G and B channels used in Q2.1.

Q2.2. Y was kept and both chroma dimensions were reduced by factors 2, 4 and 8, using area reduction and linear reconstruction. Factor 2 changes $3N$ samples to $N + N/4 + N/4 = 1.5N$: 786,432 to 393,216 bytes, saving 50%. Factors 4/8 need 294,912/270,336 bytes, saving 62.5/65.625%. These are stored channel samples before display reconstruction, not compressed files. Factor 2 looked close; factor 4 softened colour boundaries and factor 8 showed clearer blending, while Y detail remained.

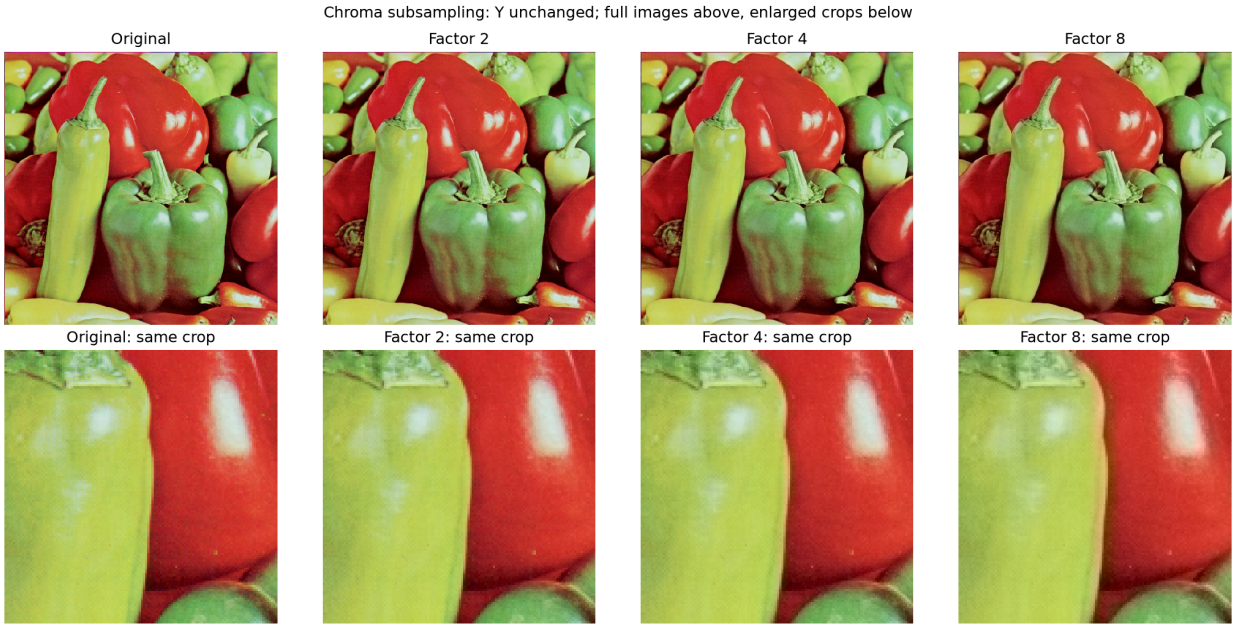


Figure 2: Chroma subsampling at factors 2, 4 and 8 compared with the original image. Enlarged crops show increasing colour smoothing while luminance detail is preserved.

3.3 Quantization and Sampling

Q3.1. RGB integer division by 16 gave 16 possible levels per channel, displayed as $0, 16, \ldots, 240$. Only 15 were occupied because no input reached 240; smooth colours became slightly stepped and darker. Packed 4-bit

storage would give 2:1 reduction, or 50% saving: 393,216 versus 786,432 bytes. The actual `uint8` array still occupied 786,432 bytes.

Q3.2. Grayscale division by 32 grouped 32 input intensities into each of eight levels, displayed as $0, 32, \ldots, 224$. All eight were occupied; the last held only 29 pixels. Both histograms counted 262,144 pixels. Combining nearby intensities caused banding and lost shading and texture, although outlines remained clear (Figure 3).

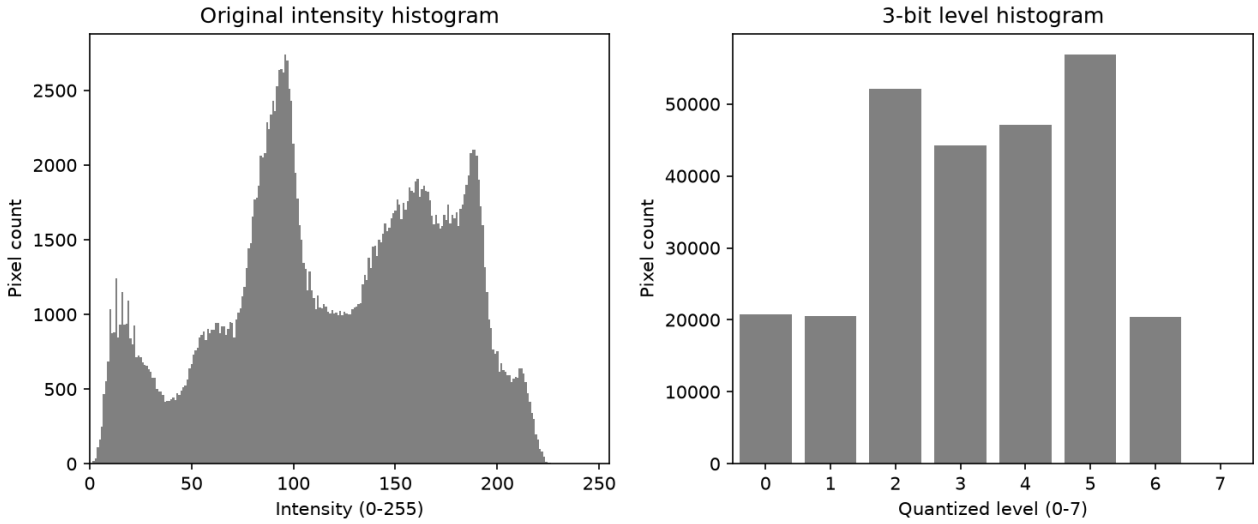


Figure 3: Q3.2: original intensities (left) and 3-bit levels (right), using separate count scales.

3.4 Convolution and Edge Detection

Q4.1. Matching Gaussian kernels were compared at $\sigma = 1$, using float64 output and `BORDER_REFLECT_101`, after warm-up (Table 2). Separable filtering was faster only at 5×5; ordinary 2D was slightly faster at 11 and 21. Costs fall from k^2 to about $2k$ weights per pixel, potentially helping large kernels, but library implementation affects real speed. Both outputs looked alike, differing by at most about 0.000002 intensity units.

Table 2: Recorded average times; loading and plotting excluded.

Task (first / second method)	Runs	First (ms)	Second (ms)
Gaussian 5×5 (2D / separable)	30	5.420	1.429
Gaussian 11×11 (2D / separable)	30	9.880	10.455
Gaussian 21×21 (2D / separable)	30	23.481	24.022
Brightness (NumPy / loops)	5	2.040	468.915
Log transform (direct / lookup)	30	9.469	2.468

Q4.2. The 3×3 Sobel function used `dx=0, dy=1`. Absolute top-to-bottom brightness changes highlighted horizontal boundaries and weakened long vertical sides. With `dx=1, dy=0`, vertical edges stood out; curved edges appeared in both (Figure 4).

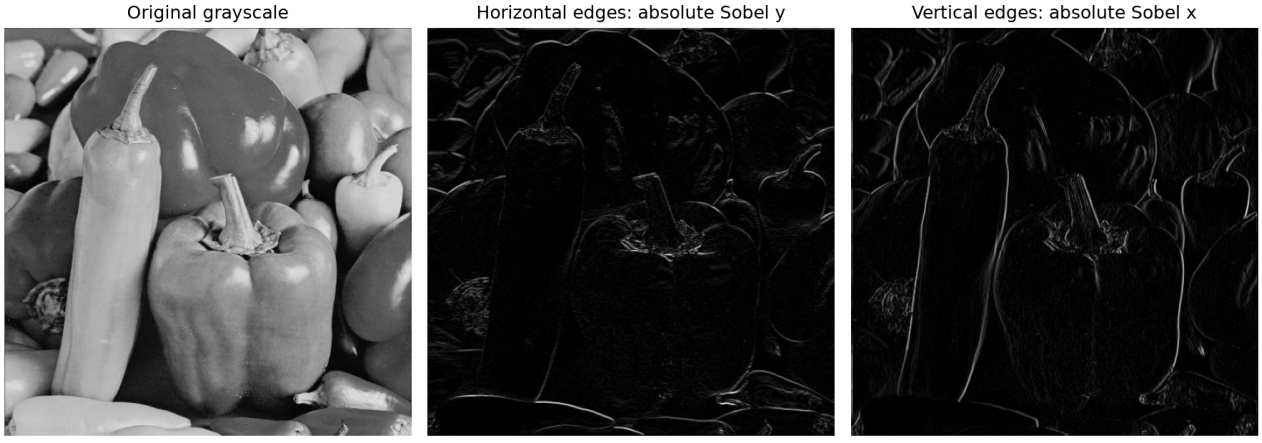


Figure 4: Q4.2: Sobel edge detection. Horizontal edges use `dy=1`; vertical edges use `dx=1`.

Q4.3. Signed derivatives combined as $G = \sqrt{G_x^2 + G_y^2}$; a copy was normalized to 0–255. Both directions gave more complete outlines and stems. Brighter pixels mean stronger edges (Figure 5).

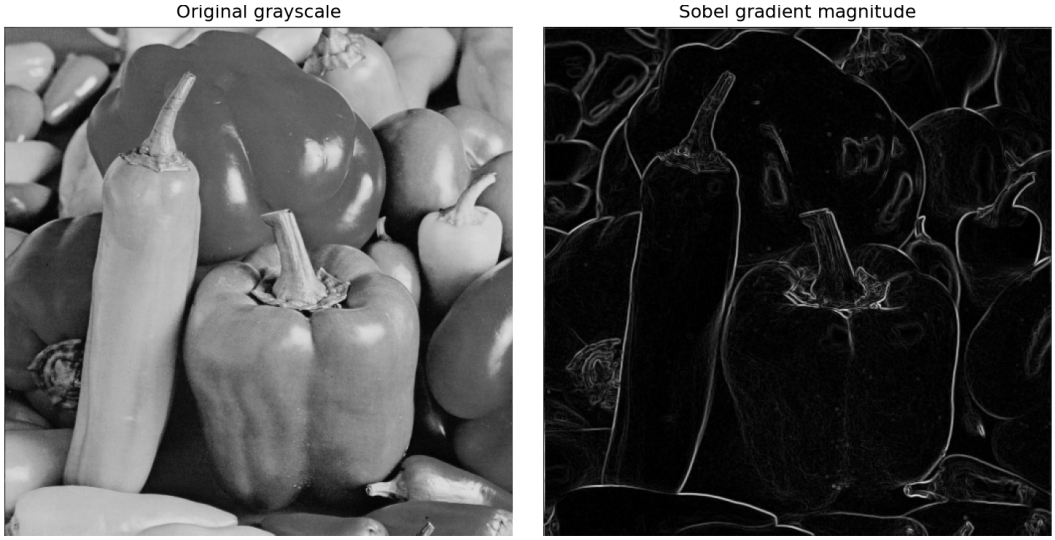


Figure 5: Q4.3: Original grayscale image and Sobel gradient magnitude.

3.5 Matrix Operations

Q5.1. Stacking pixels, applying $1.2(I - 128) + 128 + 20$, clipping to 0–255 and reshaping gave identical brighter images from NumPy and loops, with clipped highlights. NumPy was 229.82 times faster from unrounded times (Table 2), avoiding per-pixel Python operations. Timings included calculation, output creation and conversion.

Q5.2. The stacked image has 262,144 values, so H has $262{,}144^2 = 68{,}719{,}476{,}736$ elements. Float32 needs 274,877,906,944 bytes (256 GiB); float64 needs 549,755,813,888 bytes (512 GiB), where GiB is 1024^3 bytes. H was not allocated. Most entries would be zero; keeping nine kernel weights and processing local neighbourhoods is more practical.

3.6 Pixel-Based Processing

Q6.1. Direct and 256-entry lookup versions gave identical `uint8` images, range 0–240. Lookup was 3.84 times faster (Table 2) because it reused results instead of recalculating logarithms. Its separate 0.539 ms construction was excluded; applications were warmed up and averaged over 30 calls.

Q6.2. Two log halves joined at 127.5, strength 4 (Figure 6, right). The curve is steeper centrally and flatter at both ends: input gaps of 32 gave output gaps 17 (0–32), 64 (112–144), and 18 (223–255). Middle shading separated more; shadow/highlight differences compressed. Dark gaps looked darker and light areas brighter. Intensity order and endpoints 0 and 255 were preserved.

3.7 Elementary Images

Q7.1. Outer products of the teacher's vectors made 16×16 images. Ramp×Ramp gave smooth two-direction changes; Step×Ramp a boundary plus a gradient; Step×Step a sharp lower-right block; Step×Sine repeating columns in the lower half. The shared -1 to 1 scale made zero middle gray. See Figure 7.

Q7.2. Least squares fitted Step×Ramp, Step×Step and Step×Sine to the 8×8 square (rows 8–15, columns 4–11, zero-based). Weights were 0.156058, 0.812731 and 0.688038; MSE 0.107400 on 0–1, raw range 0.000000–0.895961. The horizontal boundary matched, but interiors were uneven, side edges wrong and bands remained outside the square. Three patterns gave a coarse fit.

3.8 Transforms and Frequency Domain

Q8.1. The display used `log1p(abs(fftshift(fft2(gray))))`. The bright centre held average brightness and slow changes; faster changes lay farther out. Horizontal/vertical streaks included edges and border effects: vertical edges contribute along horizontal frequency and vice versa. Curves involve several directions. The FFT does not locate individual edges. See Figure 8.

Q8.2. A circular low-pass mask kept radii below 40 bins in the 512×512 grid; inverse shifting and FFT returned its real part. Main shapes remained, detail weakened and boundaries softened, with ringing from the sharp cutoff. Raw values were -17.078 to 243.781; negatives displayed black without changing the data.

Q8.3. Floating-point original-minus-low-pass kept outlines, stems and detail around near-zero smooth regions. The raw range was -155.400 to 113.396; display limits were ± 155.400, with zero middle gray. Ringing remained.

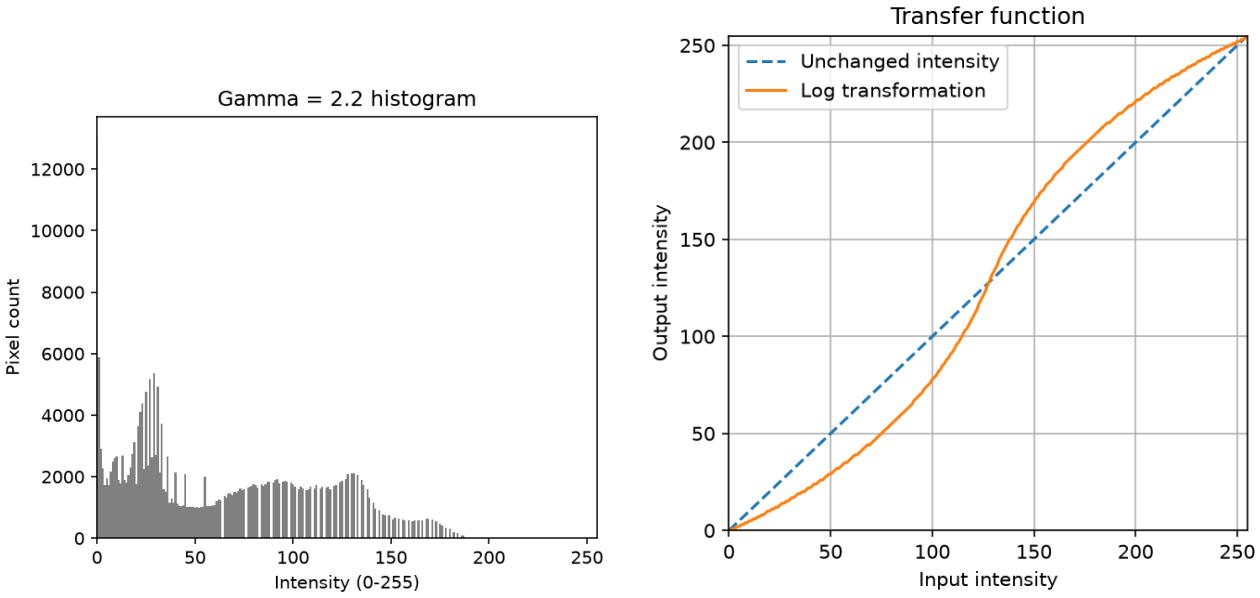


Figure 6: Left: gamma-2.2 histogram, compared with Figure 3's original using its labelled count scale. Right: Q6.2 logarithmic transfer function; the dashed line leaves intensity unchanged.

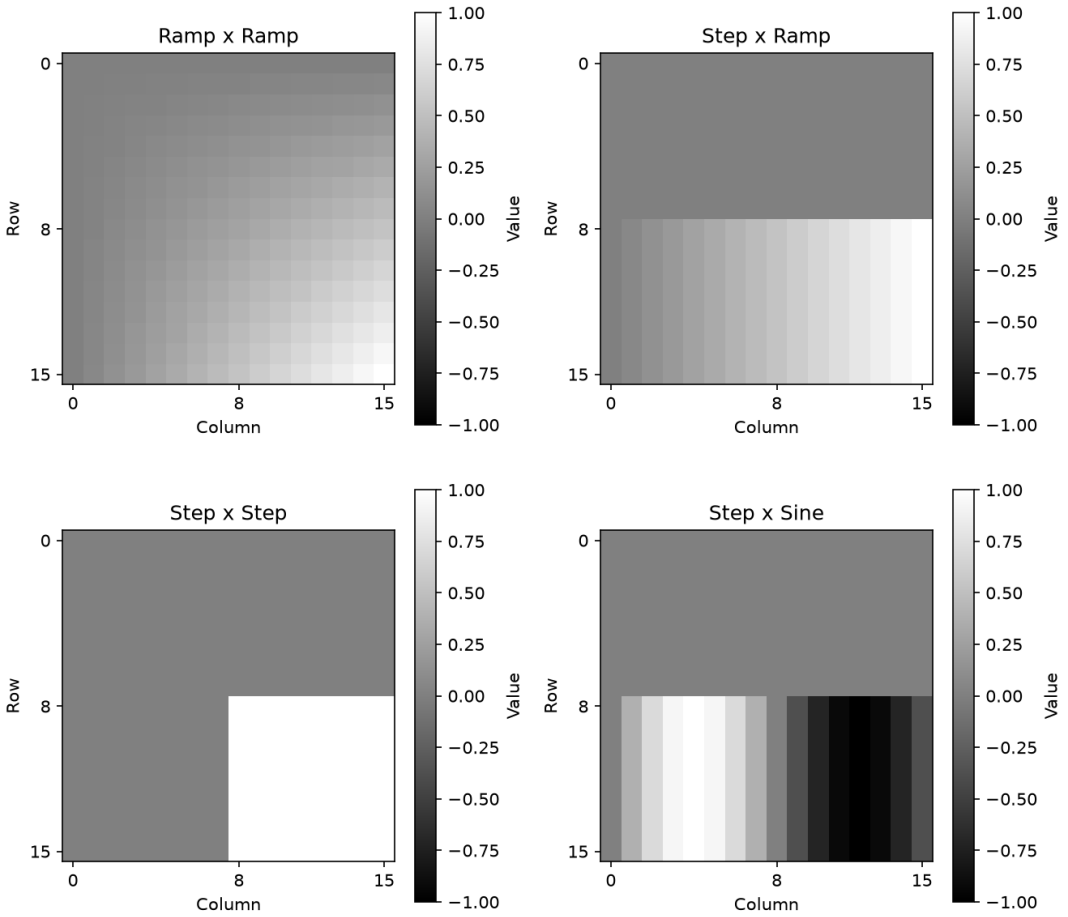


Figure 7: Q7.1: Four elementary 16×16 images generated from combinations of ramp, step and sine basis vectors.

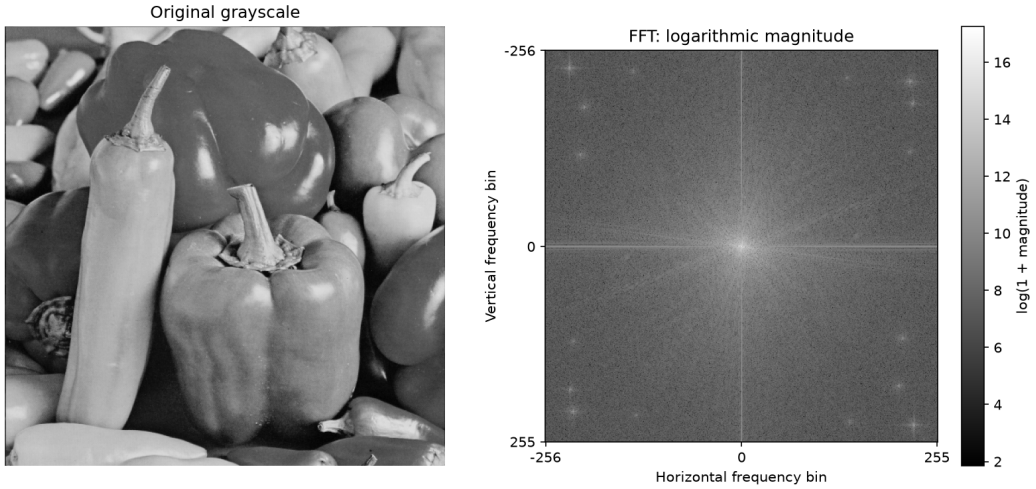


Figure 8: Q8.1: Original grayscale image and its centered 2D FFT magnitude spectrum (log scale).

Low-pass plus high-pass recovered the original with maximum error 0; the inverse FFT's largest imaginary part was 6.172×10^{-14}. See Figure 9.

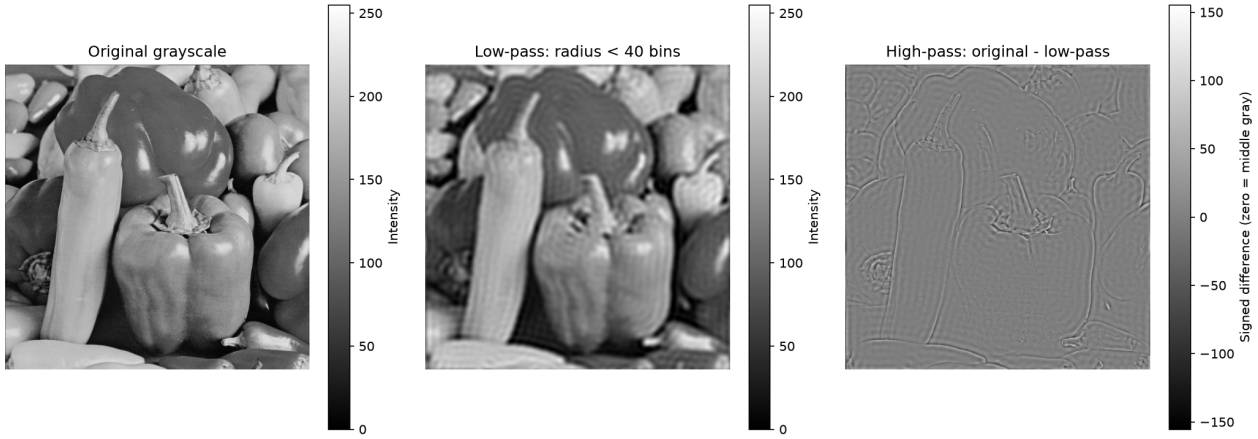


Figure 9: Q8.2–8.3: Original grayscale image, low-pass result (radius below 40 bins), and signed high-pass difference. Zero is middle gray in the high-pass image.

3.9 Histograms and Image Enhancement

Q9.1. Figure 10 shows that pixel intensities are concentrated mainly in the darker and middle ranges. The largest histogram peak was at intensity 96, and 76.43% of pixels were between 64 and 191. This indicates that the image contains mostly midtones and darker areas, with relatively few bright pixels. The cumulative histogram confirms this: 52.51% of pixels had intensities at or below 127, and 92.19% were at or below 191. It reached 100% at intensity 228. The histograms describe how brightness is distributed, but not where individual objects or details appear in the image.

Q9.2. Reducing contrast with $128 + 0.5(I - 128)$ compressed the grayscale range from 0–228 to 64–178, making the image flatter and reducing differences between dark and light regions. Histogram equalization using $T(r) = 255\,\mathrm{CDF}(r)$ expanded the intensity range to 0–255. Compared with the reduced-contrast image, shading and details became clearer. Compared with the original, dark and light regions showed stronger contrast, although grain became more visible (Figure 11). The custom CDF method and OpenCV's `equalizeHist()` produced nearly identical results, with a maximum difference of 1 intensity level and a mean absolute difference of 0.443756. The small differences arise from OpenCV's cumulative-histogram normalization and rounding, compared with the simpler implemented transformation.

3.10 Noise and Noise Reduction

Q10.1. With seed 42, one zero-mean Gaussian sample was scaled and added in floating point, then clipped before integer conversion. Median 3×3 at noise $\sigma = 25$ changed PSNR from 20.318 to 26.829 dB; bilateral (diameter 9, colour/spatial sigmas 75/75) at $\sigma = 35$ changed 17.519 to 26.888 dB. The 0.059 dB gap closely matched outputs, but different input noise prevents a same-input ranking. Bilateral trials at $\sigma = 25, 30, 35$ gave 28.860, 28.004, 26.888 dB. Both reduced grain; bilateral looked smoother but softened stems. Similar PSNR did not mean identical appearance.

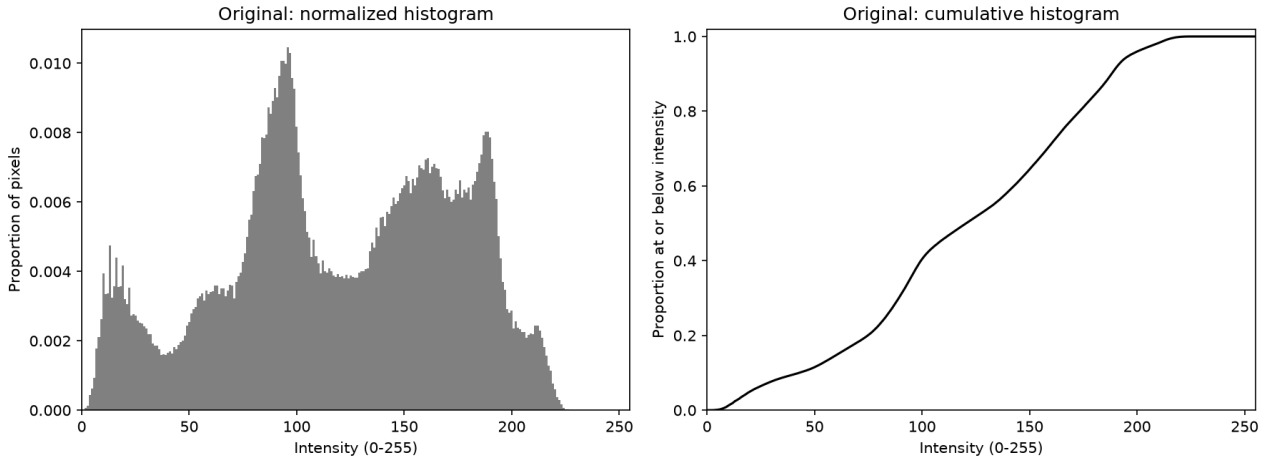


Figure 10: Q9.1: original normalized histogram and cumulative histogram (CDF).

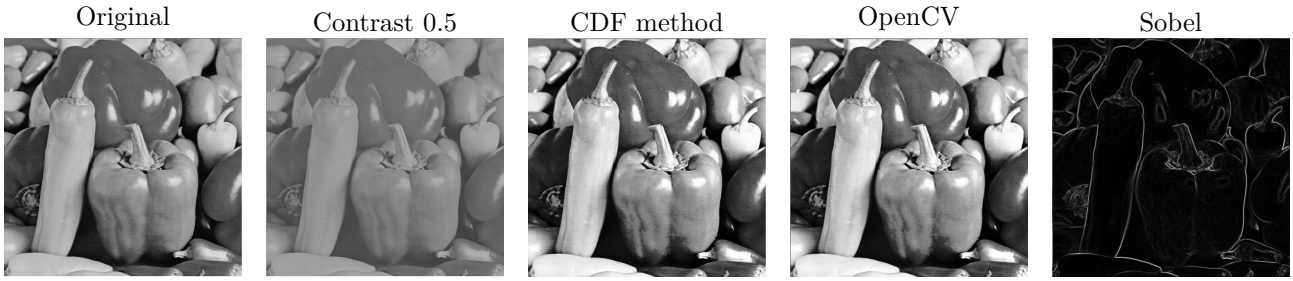


Figure 11: Q9.2: contrast/equalization (left four); Q4.3: normalized Sobel (right). Displays: 0–255.

Q10.2. Both 3×3 filters received the same noisy image for each row of Table 3. Salt-and-pepper probability was 4% total (2% black, 2% white without overlap); 3.986% of pixels were selected. Mean filtering gave slightly better Gaussian results but blurred edges. Median removed isolated extremes much better; mean filtering spread them into gray marks. `cv2.PSNR()` used the clean image, with differences from unrounded values.

Table 3: Same-input filtering: PSNR and mean-minus-median difference, all in dB.

Noise	Input	Mean	Median	Difference
Gaussian, $\sigma = 25$	20.318	27.237	26.829	+0.409
Salt-and-pepper, 4%	19.296	26.516	34.035	−7.519

3.11 Morphological Filtering

Q11.1. Two 128×128 binary tests used separate shapes. A had a circle, rectangle and triangle; B added a second circle of a different size. Each had three white specks and a 2×2 black hole. Opening with a 3×3 cross removed specks; closing with a 5×5 ellipse filled the hole. Eight-connected labels minus background gave expected/detected counts 3/3 and 4/4. Corners rounded slightly; touching objects would count together.

Q11.2. Dilation minus erosion with the cross showed pepper boundaries and more grain; Sobel outlines were smoother. Raw ranges were 0–204 and 0–812.857, separately normalized to 0–255 for display, so displayed brightness does not compare raw strength. Morphology suits binary shapes and kernel-dependent boundaries; Sobel suits grayscale brightness changes and directional derivatives. Its magnitude shows strength, not direction.

4 Discussion

Vectorization and lookup saved time with identical outputs. Gaussian timings depended on routines, not just operation counts; one run cannot give a general ranking. Filters traded noise against detail and depended on noise type. Equalization also exposed grain. The coarse shape fit and touching-object limit show why settings and representation matter.

5 Conclusion

The experiments linked pixel values, colour, detail and frequency content. Numerical results supported the image comparisons and explained time, memory and noise trade-offs.

Appendix: Code Snippets

FFT filtering. The circular mask keeps low frequencies; the inverse FFT returns the smooth image.

```
filtered_fft = fshift * lowpass_mask
lowpass_complex = np.fft.ifft2(np.fft.ifftshift(filtered_fft))
```